

Name:

Scholar No. :

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End Term Examination (Session: July-December 2024)

B.Tech. **Semester - III** **Branch: CE, Chem, EE, ECE, Mech & MME**
Subject: Mathematics-III **Code: MTH-231** **Max. Marks: 50** **Time: 120 Minutes**

Note:

- All questions are compulsory and carry equal marks. Usual notations are used.
 - All solutions should be correct to four decimal places.
 - Assume missing data if any.
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Q1(a) Using Newton Raphson method, find the real root of $x \log_{10}(x) = 1.2$ correct to four decimal places.

Q1(b) Solve the following system of linear equations, correct to four decimal places using Jacobi iteration method by taking the initial solution as $(0.5, 0.5, 0.5)$:

$$x + y + 5z = -1$$

$$5x - y + z = 10$$

$$2x + 4y = 12$$

Q2(a) Apply Newton's Divided Difference method to find the interpolating polynomial and its value at $x = 6$.

x	-4	-1	0	2	5
y	1245	33	5	9	1335

Q2(b) Use the appropriate method to find the value of x when $y = 0.390$ for the following data.

x	20	25	30	35
y	0.342	0.423	0.500	0.650

Q3(a) The following data gives the height of a hot air balloon, y (in metres) at different times t (in seconds). Find the vertical velocity and the acceleration of the hot air balloon correct to four decimal places, using Stirling's formula at time $t = 900$ seconds.

t	0	300	600	900	1200	1500	1800
y	135	149	157	183	201	205	193

P.T.O.

Q3(b) Evaluate the integral $\int_0^6 \frac{dx}{1+x^2}$ by taking six subintervals, using

(i). Trapezoidal Rule

(ii). Simpson's $\frac{1}{3}$ rd Rule

Also, compute the error in each formula and compare your results. Which method gives a better answer?

Q4(a) Use modified Euler's method to estimate the value of y at $x = 2.2$ correct to four decimal places, by taking step size 0.1, for the initial value problem $\frac{dy}{dx} = -xy^2$, $y(2) = 1$.

Q4(b) Solve the differential equation $\frac{dy}{dx} = \frac{2xy + e^x}{x^2 + xe^x}$, $y(1) = 0$ using R-K method of fourth order to find $y(1.4)$ correct to four decimal places, taking step size 0.2.

Q5(a) After investigation it has been found that the demand for automobiles in a city depends mainly on the number of families in the city. The given data lists the results of a survey conducted by a consulting agency, where x represents the number of families (in lakhs) and y represents the sales of automobiles (in thousands)

x	70	75	80	60	90
y	25.2	28.6	30.2	22.3	35.4

(i). Find the correlation coefficient for the given data.

(ii). Describe the significance of the correlation coefficient obtained in your result.

(iii). Find the line of regression of y on x and hence estimate the sale for a city which has 100 lakh families.

Q5(b) By the method of least squares, fit a second degree curve to the given data.

x	1.0	1.5	2.0	2.5	3.0	3.5	4.0
y	1.1	1.3	1.6	2.0	2.7	3.4	4.1

**** End of the question paper ****